

Geography Ch-3 — Vulcanism and Earthquakes

Physical Geography | Endogenic Forces | GS-I & GS-III (Disaster)

FOUNDATION: G.C. Leong · STANDARD: Majid Husain (Indian & World Geography)

Register-style revision notes · grounded in source books

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

FOUNDATION — Volcanoes, Earthquakes & their Distribution (G.C. Leong)

GS-I / GS-III ·

Physical

Geography →

Endogenic Forces

(Source: G.C.

Leong, Certificate

Physical &

Human

Geography)

■ NEWS TRIGGER

Back-to-back tremors at Jhajjar, Haryana (M4.4 on 10 Jul 2025, M3.7 on 11 Jul 2025) shook Delhi-NCR; a M7.1 quake near the Nepal-Tibet border (7 Jan 2025) was felt across north India — Delhi lies in Seismic Zone IV (high risk).

■ INTRO & ORIGIN

Vulcanism and earthquakes are **endogenic** (internal) processes. Volcanoes are vents through which magma, gases and fragments reach the surface; earthquakes are ground vibrations from sudden crustal movement.

*By activity, Leong groups volcanoes into **active, dormant and extinct** — and every volcano passes through these stages in its life-cycle. Their global spread mirrors that of earthquakes.*

■ KEY DATA

Category / Form	Basis	Feature / example
Active	Erupts frequently / recently	e.g. Pacific-rim cones
Dormant	Erupted before; may erupt again	'Sleeping' volcano
Extinct	No historic eruption; retains form	Deccan (old)
Shield / lava dome	Highly fluid basic lava	Gentle slopes; extensive lava sheets
Acid lava cone	Sticky viscous lava	Steep, short-flow cone

■ STATIC THEORY

- **Shield (lava dome) volcanoes** form from fluid basic (basaltic) lava that spreads far → gentle slopes and wide lava sheets.
- **Acid lava cones** form from viscous lava that cannot flow far → steep-sided cones.
- **Distribution of earthquakes coincides closely with volcanoes** — both cluster along the same belts.
- The **Circum-Pacific 'Pacific Ring of Fire'** is the zone of greatest seismicity — ~70% of earthquakes occur here.
- A second major belt is the **Mediterranean-Himalayan (mid-continental) belt** — Asia Minor, the Himalayas, etc.
- A third zone follows the **Mid-Atlantic ridge**; lava plateaus (Deccan, Columbia-Snake) mark past fissure vulcanism.

■ MUST-KNOW FACTS

1. Volcano life-cycle: active → dormant → extinct.
2. Shield volcano = fluid basaltic lava, gentle slope; acid cone = viscous lava, steep slope.
3. Pacific 'Ring of Fire' = ~70% of world earthquakes.
4. Second belt = Mediterranean-Himalayan (mid-continental).
5. Earthquake and volcano distributions coincide (plate margins).

■ UPSC TRAPS

- ✗ Shield volcanoes have steep sides.
- ✓ Shield volcanoes have GENTLE slopes (fluid lava); acid lava cones are steep (viscous lava).
- ✗ Extinct volcanoes have no volcanic features.
- ✓ Extinct = no eruption in historic times but they RETAIN volcanic form.
- ✗ Earthquakes and volcanoes occur in unrelated zones.

6. Delhi-NCR = Seismic Zone IV (high damage risk).

✓ Their distributions coincide — both follow plate-margin belts (Ring of Fire, Himalayan belt).

■ MAINS ANGLE

Link the Himalayan seismic belt to India's disaster vulnerability (Zone IV-V) — a bridge from physical geography to disaster-management (GS-III) answers.

■ STUDY LINK

Geography → Physical → Volcanism & Seismicity (Leong Ch-3)

HIGH

STANDARD — Volcanic Forms, Earthquake Mechanics & Plate Control (Majid Husain)

GS-I / GS-III ·
Geomorphology
→ Endogenic
Forces (Source:
Majid Husain,
Indian & World
Geography)

■ NEWS TRIGGER

Earthquake magnitude (Richter) vs intensity (Mercalli) and focus/epicentre are the exact terms the National Center for Seismology uses to report the 2025 Delhi-NCR and Nepal quakes.

■ INTRO & ORIGIN

There are ~**600 active volcanoes** in the world, most in the Pacific. By form, Majid Husain adds the **composite volcano** (alternating ash + lava → high, steep cones) and the **caldera** (collapsed summit, may fill as a crater lake).

Eruption style varies with lava: **Icelandic** (quiet fissure) and **Hawaiian** (fluid, quiet) at one end; explosive at subduction zones. Volcanoes concentrate along **plate margins** — subduction of oceanic-continental (Mt St Helens) or oceanic-oceanic (Japan, Philippines).

■ KEY DATA

Earthquake term	Definition	Note
Focus (hypocentre)	Point of origin inside the Earth	Most foci < 60 km deep
Epicentre	Point on surface directly above focus	First to feel shaking
Richter scale	MAGNITUDE (energy released)	Logarithmic; +1 = 10× amplitude, ~31.5× energy
Mercalli scale	INTENSITY (surface damage felt)	Descriptive, place-specific
Seismograph	Instrument recording waves	~5,00,000 quakes/yr recorded

■ STATIC THEORY

- **Composite volcanoes** (stratovolcanoes) are built of alternating ash and lava layers → high, steep, symmetrical cones.
- A **caldera** forms when a volcanic summit collapses inward after magma loss; rainwater can fill it as a **crater lake**.
- **Earthquake mechanics:** rocks strained beyond their elastic limit rupture along active faults, suddenly releasing stored energy as seismic waves radiating from the focus.
- The Earth's crust has **7 major + several minor plates** in constant relative motion; plate movement is the primary cause of earthquakes.
- **Richter is open-ended & logarithmic** — each whole number = 10-fold rise in wave amplitude and ~31.5-fold rise in energy; magnitude ≠ intensity (Mercalli).
- Volcano distribution is tightly tied to plate margins (subduction & spreading) — hence the Ring of Fire concentration.

■ MUST-KNOW FACTS

■ UPSC TRAPS

1. ~600 active volcanoes worldwide, mostly Pacific. 2. Composite volcano = alternating ash + lava (steep cone); caldera = collapsed summit → crater lake. 3. Focus = origin inside Earth; epicentre = surface point above it; most foci < 60 km. 4. Richter = MAGNITUDE (log, +1 ≈ 31.5× energy); Mercalli = INTENSITY. 5. Earthquakes = rock strained beyond elastic limit → rupture along faults. 6. 7 major lithospheric plates; plate tectonics is the chief cause of earthquakes.	✗ Richter and Mercalli measure the same thing. ✓ Richter = magnitude (energy released); Mercalli = intensity (damage felt at a place). ✗ Focus and epicentre are the same point. ✓ Focus is inside the Earth (origin); epicentre is the surface point directly above the focus. ✗ A one-point rise on Richter doubles the energy. ✓ Each whole number = 10× wave amplitude and ~31.5× energy — not double.
■ MAINS ANGLE <i>Analyse why the Indian-Eurasian plate collision makes the Himalaya a high-magnitude seismic belt, and how Richter/Mercalli data drive seismic zonation and building codes — geomorphology into governance.</i>	■ STUDY LINK Geography → Geomorphology → Endogenic forces & Plate Tectonics (Majid Husain)

End of Register Notes — UPSC Agent / Copilot CLI